



# Frontal-occipital phase synchronization predicts occipital alpha power in perceptual decision-making

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Received: 3 February 2022 / Revised: 19 June 2022 / Accepted: 21 July 2022 / Published online: 24 August 2022  
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## Abstract

Numerous studies of perceptual decision-making have shown that lower prestimulus alpha power leads to a higher hit rate in visual detection, which is believed to correlate with the top-down control. However, whether frontal-occipital phase synchronization underlying the top-down control could impact the occipital alpha power that directly affects the perceptual performance remains unclear. In this study, we used analyses of the general linear mixed model (GLMM) and event-related potentials (ERPs) to show that the prestimulus alpha power over the occipital area directly affected visual perception. Using both the univariate and multivariate methods, we found that low-frequency (4–30 Hz) frontal-occipital phase synchronization predicted the prestimulus alpha power over the occipital area. Overall, our results suggested that frontal-occipital phase synchronization could predict occipital alpha power that directly affects perceptual decision-making.

**Keywords** Phase synchronization · Prestimulus alpha power · Perceptual decision-making

## Introduction

Humans make numerous decisions based on sensory information in everyday life. This cognitive process could be defined as perceptual decision-making that humans detect, discriminate, or categorize information obtained from senses (Hanks and Summerfield 2017; O’Connell and Kelly 2021). In perceptual decision-making studies, researchers have put much attention on the prestimulus alpha-band neural oscillations, given that alpha oscillations reflecting

neural excitability (Iemi et al. 2021; Lange et al. 2013; Romei, Brodbeck, et al., 2008; Romei, Rihs, et al., 2008) would affect perceptual decision-making performance significantly (Iemi and Busch 2018; Samaha et al. 2020). A good example comes from the finding that lower prestimulus alpha-band power leads to a higher hit rate (more ‘I saw the stimulus’ responses) for near-threshold stimuli in visual detection tasks (Brüers and VanRullen 2018; Busch et al. 2009; Busch and VanRullen 2010; Ergenoglu et al. 2004; Iemi et al. 2017; Iemi and Busch 2018; Mathewson et al. 2009, 2014; Limbach and Corballis 2016; Van Dijk et al. 2008). Despite the high replicability of this finding, the issue of whether frontal-occipital phase synchronization underlying the top-down control could impact the occipital alpha power that directly affects the visual perception remains elusive.

Previous studies have shown that the prestimulus alpha-band power over many brain areas, not just the occipital area, could predict the subsequent visual perception. Prestimulus alpha power over brain areas, including occipital (Van Dijk et al. 2008), parietal (Iemi et al. 2017; Mazaheri et al. 2009), and frontal (Busch et al. 2009), were separately found to lead to perceptual differences. This inconsistency obscured the prestimulus neural process. For the occipital activity predicting the visual perception, it has been indicated that occipital alpha power reflects the excitability, which affects

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the perceptual bias; namely, subjects are more likely to react to whether signal or noise (Samaha et al. 2020). The frontal activity predicting the perceptual performance implied that higher-order cognition, such as attention (Mathewson et al. 2014), participated in the prestimulus stage of perceptual decision-making (Busch and VanRullen 2010). However, little attention has been paid to the interaction of different brain areas in the prestimulus stage. Visual perception involves both top-down and bottom-up processes (Gilbert and Li 2013). For instance, previous studies suggested that occipital excitability was modulated by top-down control from frontal areas (Busch and VanRullen 2010; Mathewson et al. 2014; Diepen et al. 2017; Veniero et al. 2021; Zhang et al. 2019). Therefore, the activity over many brain areas may predict visual perception with different spatiotemporal patterns. It was likely that the alpha power over the occipital area directly influences the visual perception while the alpha activity of other brain areas indirectly influences the perception by impacting the occipital alpha power through top-down control in the prestimulus stage of the perceptual decision-making.

If the frontal activity influenced the occipital prestimulus alpha power, how did the underlying top-down control take effect? One candidate mechanism was the long-range phase synchronization, which refers to the information transfer between two brain areas via oscillation phases (Noudoost et al. 2010; Fell and Axmacher 2011; Clayton et al. 2015). Previous studies have shown that phase synchronization, which underlies top-down control, plays a critical role in many cognitive functions, such as memory and attention (Noudoost et al. 2010; Fell and Axmacher 2011; Clayton et al. 2015). The long-range phase synchronization also affected visual perception. It was indicated that synchronous oscillations in the alpha frequency band suppressed the perception of shortly presented stimuli while oscillational synchrony in higher frequency bands boosted visual perception (Hanslmayr et al. 2007). In addition, Domenica et al. discovered that both visual perception and visual cortex excitability could be modulated by the top-down control from the frontal eye fields (FEF) through beta-band phase realignment in a motion discrimination task (Veniero et al. 2021). However, there were some issues regarding frequency bands, experimental designs, etc. For example, beta-band modulation was identified (Veniero et al. 2021) despite the prominence of alpha-band activity in visual perception (Samaha et al. 2020; Hanslmayr et al. 2007). Also, it was unclear that consistent results could be yielded across closely related experimental designs, such as visual detection and discrimination tasks (Samaha et al. 2020). Therefore, it was worth investigating further whether frontal-occipital phase synchronization in various frequency bands affects occipital

alpha power in the prestimulus stage of the visual detection tasks, which remains unexamined so far.

This study aimed to investigate whether frontal-occipital phase synchronization underlying the top-down control could impact the occipital alpha power that directly affects visual perception. To this end, we conducted a visual detection experiment and analyzed multiple EEG signatures. When the cluster-based permutation test was used to tentatively identify the distribution of alpha power, a more precise spatiotemporal distribution of global alpha power was yielded by the general linear mixed model (GLMM). Event-related potentials (ERPs) over the occipital area were compared between hit and miss trials to convince further that it was the occipital area where the perceptual performance differentiates. After determining that the occipital alpha power directly impacts the perceptual behavior, we first compared the phase synchronization of different frequency bands in a univariate way between low- and high-alpha-power conditions. Then we used the time series of phase differences to predict the occipital alpha power in the prestimulus stage across subjects in a multivariate way by using a long short-term memory (LSTM) classifier.

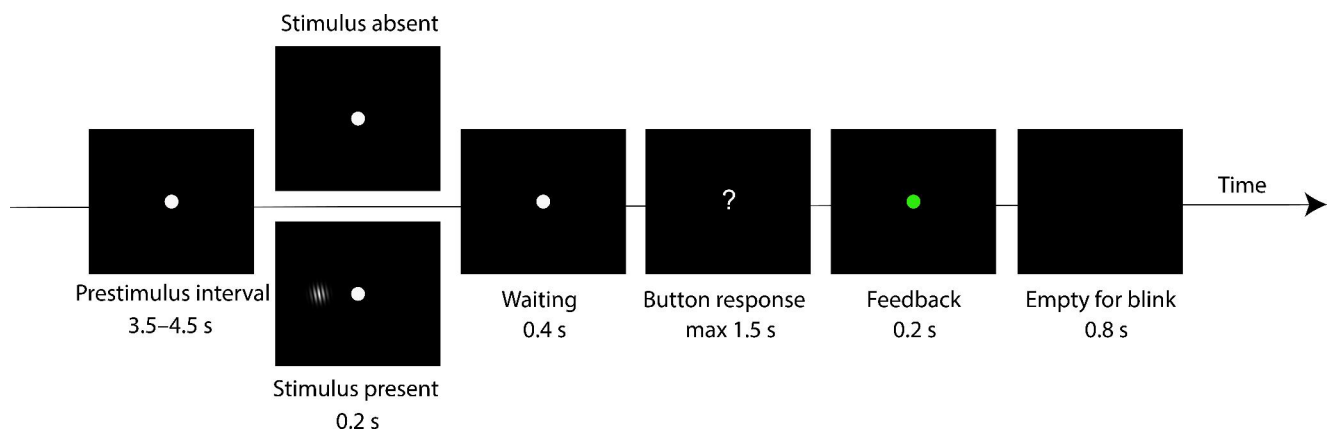
## Materials and methods

### Participants

A total of 30 volunteers (17 males, mean age  $22.2 \pm 1.5$  years) participated in this study. All participants were on-campus students and had normal or corrected-to-normal vision. None of them suffered from neurological or psychological disorders or took medication that interfered with the study. Before formal participation, they were required to rest well and refrain from consuming caffeine or alcohol. Participants provided informed consent before the start of the experiment. The experiment was approved by the local ethics committee at Shenzhen University.

### Stimuli and paradigm

We adopted similar experimental settings in the studies (Iemi et al. 2017; Iemi and Busch 2018). Participants were seated in a dark soundproof room. Experimental stimuli were presented on a black background of a 21-inch LCD monitor operated at 100 Hz. The monitor was positioned at 60 cm in front of the participant. The experimental paradigm was adapted from previous studies (Iemi et al. 2017; Iemi and Busch 2018) and coded in MATLAB (The MathWorks, Inc., Natick, MA, US) using the Psychophysics Toolbox 3 (<http://psychtoolbox.org>).



**Fig. 1** Schematic illustration of trials. A near-threshold visual stimulus was shown on the left or right side (probability: 50%) of the fixation dot for 0.2 s in 60% of trials. After waiting for 0.4 s, participants needed to respond with buttons according to whether seeing the stimulus when the fixation dot turned into a question mark. The green (correct) or red (incorrect) dot acted as the feedback and was shown for 0.2 s. A blank screen was presented for participants to blink. The next trial started after 3.5–4.5 s. There were a total of 16 blocks. Each block consisted of 40 trials, followed by a break lasting 1.5–3 min

Participants were told to focus their gaze on the center area around the fixation dot throughout the experiment. They were required to report whether they saw the target stimuli, which consisted of low-contrast Gabor patches tilted 10 degrees with a visual angle of 0.75 degrees. The stimuli were displayed randomly at the fixation center's left or right with a visual angle of 10 degrees (Fig. 1). The stimuli side and tilt were counterbalanced across trials and did not affect the response type. The target stimuli were present in 60% of the trials and absent in 40% of the trials. The fixation cross became a question mark after 400 ms when the stimulus disappeared. The participants had to press a button for any response (see or not see). After the decision-making, participants were shown color-coded feedback: green means correct; red means incorrect. There were 16 blocks of 40 trials each. The participants were told that some trials contained a target stimulus and some did not. On each trial, an adaptive staircase procedure (QUEST) was used to determine a stimulus contrast which yielded a hit rate of 50% (Iemi et al. 2017).

### Behavioral analysis

To see whether near-threshold stimuli were given effectively to participants, trial numbers and proportions of hit, miss, correct rejection (CR), and false alarm (FA) were calculated within and across participants. Conditions of hit and miss were compared directly to see whether they had similar proportions.

### EEG recording and preprocessing

EEG signals were acquired from a 64-channel EEG electrode system (Easycap) and an EEG amplifier (BrainAmp,

Brain Products GmbH, Germany) according to the international 10–10 system. Electrode impedances were kept below 10 k $\Omega$  during the whole experiment. The reference and ground electrodes were FCz and AFz, respectively. The complete list of EEG sensors can be found in Fig. S1 in supplemental data.

After recording, raw EEG signals were preprocessed by EEGLAB toolbox (Delorme and Makeig 2004). They were cleaned from apparent artifacts by visual inspection, band-pass filtered from 1 to 40 Hz using FIR filters, and re-referenced offline to the average of both left and right mastoid sites. Then, the independent component analysis (ICA) was done to separate eye movements, eye blinks, and other noise artifacts. The independent components were visually inspected, and those representing artifacts were removed.

### Time-frequency decomposition

The time-frequency power of all trials was computed by custom-written MATLAB scripts. EEG data were segmented into epochs from –1500 ms to 500 ms relative to the onset of the stimulus. Single-trial epochs were decomposed into their time-frequency representations via convolution with a family of complex Morlex wavelets, defined as Gaussian-tapered complex sine waves. Thirty linearly spaced frequencies between 1 and 30 Hz were utilized, and the number of cycles for each frequency increased linearly from 1 to 12 (Iemi et al. 2017). The convolution was conducted through frequency-domain multiplication, in which the Fourier-derived spectrum of epochs was multiplied by the spectrum of wavelets. Then the inverse Fourier transform was performed. Power was extracted as the squared magnitude of the complex result. Note that the power had no baseline correction because we mainly focused on

prestimulus oscillations. Also, we mainly analyzed time-frequency points before  $-150$  ms considering the windowing effect (post-stimulus data were “leaked” to prestimulus interval) (Iemi et al. 2017).

### Cluster-based permutation test

To investigate how prestimulus alpha power impacts behavior performance, we adopted two typical strategies that previous studies used to implement (Iemi et al. 2017; Limbach and Corballis 2016). In the first strategy, trials corresponding to hit and miss were extracted (Limbach and Corballis 2016), and a cluster-based permutation test (Maris and Oostenveld 2007) was conducted in the electrode space in the alpha band (8–13 Hz). To perform a cluster-based permutation test, power data in two conditions (hit and miss) were shuffled in each subject 3000 times. The random data were then used to establish null distributions of effects under the existing bias. Next, clusters of spatially adjacent significant differences (threshold  $p < 0.05$ ) were identified. The sum of  $t$ -values in each cluster was calculated in the original data and the permuted data. If no  $t$ -value reached significance in one of the permutations, a cluster value of 0 was assigned. The significance of clusters was assessed by calculating the rank of the cluster  $t$ -values in the distribution of random data. A cluster was interpreted as significant if a higher cluster  $t$ -value was found in less than 5% of the random permutations. The cluster-based permutation test was conducted through the Fieldtrip toolbox (Oostenveld et al. 2011). In the second strategy, for each time, frequency, electrode point in each subject, power values were rearranged from small to large across trials (Iemi et al. 2017). Rearranged trials were divided into five bins, with the first and last bins representing low and high power, respectively. Hit rates were then calculated within the first and last bins. That is, the hit rate in the time-frequency domain was computed in two conditions: low- and high- power. The final step was to use a cluster-based permutation test as the first strategy did.

### General linear mixed model (GLMM)

Different from  $t$ -test and ANOVA, the general linear mixed model (GLMM) allows us to make the most of single-trial alpha power, subject, and block information (Brown 2021; Yu et al. 2021). We used GLMM to test whether hit or miss could be predicted by alpha power with specific spatiotemporal patterns, i.e., in certain brain areas and time windows. First, the whole brain areas were divided into 6 subareas: right frontal (EEG electrodes: FC4, F6, F4, F2, AF4, FC6, FC2, Fz), left frontal (EEG electrodes: FC3, F5, F3, F1, AF3, FC5, FC1, Fz), right parietal (EEG electrodes: P2, P4,

P6, CP2, CP4, CP6, C2, C4, C6, Pz, CPz, Cz), left parietal (EEG electrodes: P1, P3, P5, CP1, CP3, CP5, C1, C3, C5, Pz, CPz, Cz), right occipital (EEG electrodes: O2, PO4, PO8, Oz, POz), left occipital (EEG electrodes: O1, PO3, PO7, Oz, POz). Note that we included the central-line electrodes (e.g., Fz) in both brain hemispheres. Second, six time windows were selected, ranging from  $-650$  ms to  $-150$  ms with a step size of 100 ms. Consequently, there were  $6 \times 5 = 30$  spatiotemporal windows. In each window, GLMM was conducted using lme4 (Bates et al. 2015) package in R (Team & Al., 2013). Specifically, alpha power was taken as the fixed effect, subject number and block number were taken as the random effects, and hit and miss were taken as dependent variables. The GLMM model was specified as follows:  $\text{choice} \sim 1 + \text{alpha power} + (1|\text{block}) + (1|\text{subject})$ . Lastly, the statistical results in each spatiotemporal window were summarized in the table and estimated hit rates were plotted as regression lines.

### ERPs analysis

ERPs were analyzed to see whether the occipital area reacted differently to the near-threshold stimulus between hit and miss conditions. EEG hit and miss trials were low-pass filtered using EEGLAB filters at 30 Hz. ERPs were extracted by averaging trials in each condition (hit/miss) for each subject. The baseline correction interval was  $[-200, 0]$  ms. After these procedures, the ERPs, specifically over the occipital area (POz), were shown to demonstrate the response difference between hit and miss trials. The cluster-based permutation test was used to establish the effect latencies showing response differences (Maris and Oostenveld 2007). The topography of ERPs in  $[0, 400]$  ms was shown to see whether there was a response difference between hit and miss conditions before presenting the participant with the question marks, namely 400 ms before the stimulus onset.

### Univariate analysis method for frontal-occipital phase synchronization

Phase synchronization in EEG analysis is also known as phase-based connectivity or phase coherence (Cohen, 2014). Here we used the debiased weighted phase lag index (dWPLI) to estimate phase synchronization (Vinck et al. 2011). This index weights phase angle differences according to their distance from the real axis such that vectors closer to the real axis have a smaller influence on the estimation of connectivity. The dWPLI between two channels is calculated in Eq. 1:

$$dWPLI = \frac{\sum_{j=1}^N \sum_{k \neq j} \Im \{X_j\} \Im \{X_k\}}{\sum_{j=1}^N \sum_{k \neq j} |\Im \{X_j\} \Im \{X_k\}|} \quad (1)$$

where  $N$  is the number of trials and  $\Im\{\}$  denotes the imaginary part of the cross-spectrum  $X$  between two electrodes. Compared with other indexes, such as phase-locking value (PLV), dWPLI is not sensitive to volume conduction and more appropriate for exploratory data-driven studies. Note that the spatial filter of the surface Laplacian (Perrin et al. 1987), also known as “current source density” or “current scalp density”, was applied for all epochs before computing dWPLI. This procedure would substantially increase topographical selectivity and attenuate volume conduction, which has been demonstrated to be an essential and necessary step in the electrode-level connectivity analysis (Cohen, 2014; Cohen 2015).

Preprocessed EEG data were first band-pass filtered with [8, 13] Hz using EEGLAB filters. All trials were sorted from low to high according to the average absolute power values in [−250, −150] ms over the occipital area (POz). The first half of the trials denoted the low-power condition. The last half of the trials denoted the high-power condition. Then, phase synchronization indexed by dWPLI in [−450, −150] ms was calculated across time and averaged across low- or high-power trials. Each subject had one single dWPLI value in low and high conditions, respectively (Fig. 2a). A paired  $t$ -test was utilized to compare the averaged dWPLI in two conditions across subjects in three frequency bands: theta (4–7 Hz), alpha (8–13 Hz), and beta (4–30 Hz).

### Multivariate analysis method for frontal-occipital phase difference

Recent studies have used multivariate pattern analysis (MVPA) or “decoding” methods to understand temporal patterns of EEG. Compared to traditional univariate analysis, MVPA uses high-dimensional neural information to distinguish different experimental conditions and is often considered more sensitive (King and Dehaene 2014; Thomas and Carlson, Tijn Grootswagers 2019; Haxby et al. 2014; Ritchie et al. 2019). We here aimed to unveil whether frontal-occipital phase synchronization predicts the occipital power using the time-resolved values of phase difference. First, all trials were sorted the same way as in the univariate analysis, and two conditions were established. Second, the time series of phase differences between Fz and POz electrodes across [−450, −150] ms in each trial were extracted using the Hilbert transform. The time series of phase differences in each trial and its condition label (low or high) were taken as the input and output variables in each sample for the LSTM classifier. The reason to use LSTM was that it had been used commonly in time series classification as a variant of RNN (Pham 2021; Sherstinsky 2020). Samples were put together after being collected from all subjects; thus, we did an inter-subject classification. The sample size

was 17,443 (the total number of trials from all participants). The 20% of data was left to be the test dataset. The rest 80% was used to implement 4-fold cross-validation training. For the LSTM architecture, the number of hidden layers was 5; the number of layers was 2; the batch size was 64. The Adam optimizer was used with a learning rate of 0.001. The model was trained on an NVIDIA GeForce RTX 2070 GPU, with CUDA 11.1 using Pytorch 1.8.0 (<https://pytorch.org>). The final classification metrics included accuracy (ACC), areas under ROC curves (AUC), and F-1 score. If the ACC, AUC, and F-1 scores were all above the chance level (0.5), then the two conditions (low/high) were considered to be decoded successfully.

## Results

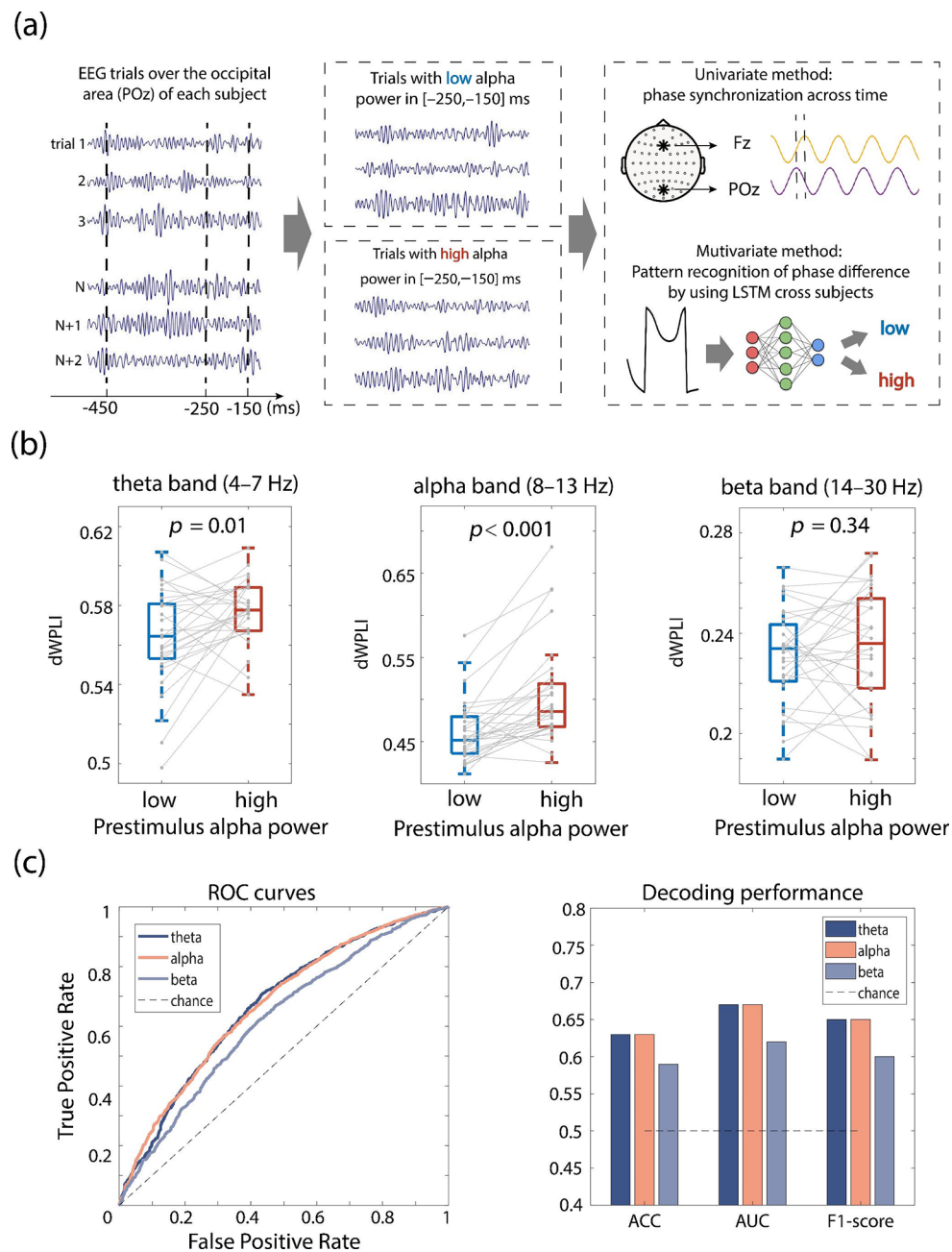
### Behavioral performance

To analyze behavioral performance, trial numbers and proportions of hit, miss, correct rejection (CR), and false alarm (FA) were compared within and across participants (Fig. 3a). The mean trial numbers and proportions within participants were 202 (31.62%), 187 (29.18%), 212 (33.24%), 35 (5.53%). Hit and miss trials had approximate proportions corresponding to the setting of near-threshold stimuli, which ideally yielded trials with the ratio of 1:1. These results indicated that our experimental setting of near-threshold stimuli was effective.

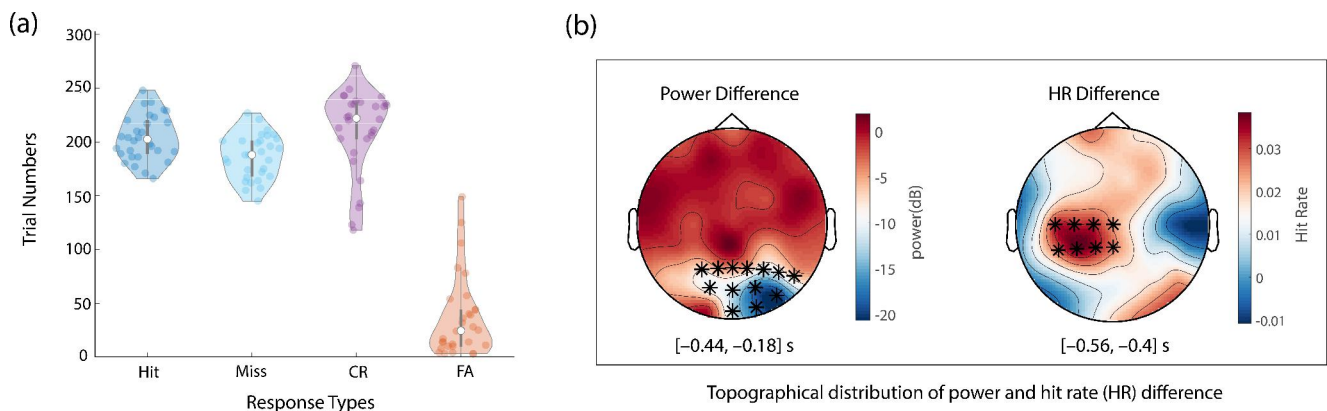
### Prestimulus alpha power could predict subsequent perceptual performance

Using cluster-based permutation tests, we employed two strategies that previous studies implemented to analyze spatiotemporal dynamics related to prestimulus alpha power. Through the first strategy, namely comparing time-frequency alpha power between hit and miss trials (Limbach and Corballis 2016), hit-trial power was significantly lower than miss-trial power over electrodes in the occipital areas in the [−0.44, −0.18] s interval ( $p < 0.01$ , Fig. 3b left panel). Through the second strategy, namely comparing the time-frequency hit rate between low- and high-power trials (Iemi et al. 2017), the low-power hit rate was significantly higher than the high-power hit rate over electrodes in the left parietal areas in the [−0.56, −0.4] s interval ( $p = 0.04$ , Fig. 3b right panel). Related brain areas covered parietal and occipital areas, in line with previous studies (Busch et al. 2009; Iemi et al. 2017; Mazaheri et al. 2009; Van Dijk et al. 2008). Given that we could not conclude that alpha power over parietal or occipital areas during those time





**Fig. 2** Phase synchronization analyses in univariate and multivariate methods. **a** Preprocessed EEG data were band-pass filtered with [8, 13] Hz. All trials were sorted from low to high according to the average absolute power values in [-250, -150] ms over the occipital area (POz). The first half of the trials denoted the low-power condition. The last half of the trials denoted the high-power condition. In the univariate method, phase synchronization indexed by dWPLI in [-450, -150] ms was first calculated across time and then averaged across low- or high-power trials. Each subject had one single dWPLI value in low and high conditions, respectively. In the multivariate method, the time series of the phase difference between Fz and Oz electrodes across [-450, -150] ms in each trial were extracted as an input sample for the LSTM classifier with the output sample of low and high labels. **b** For the univariate analysis, theta and alpha bands showed significant differences in prestimulus frontal-occipital phase synchronization between low and high conditions. **c** For the multivariate analysis, the phase difference of all three bands could be decoded by the LSTM classifier



**Fig. 3** **a** Trial numbers of hit, miss, correct response (CR), and false alarm (FA) of all participants. The hit and miss trials had approximate proportions (hit: 31.62%, miss: 29.18%), implying that near-threshold stimuli were presented to subjects effectively. **b** Left panel: Topographical distribution of power difference between hit and miss trials in the alpha band ( $[-0.44, -0.18]$  s, hit minus miss). Right panel: Topographical distribution of hit rate (HR) difference between low and high alpha power conditions ( $[-0.56, -0.4]$  s, low minus high)

intervals predicted hit rate in this visual detection task, we used GLMM to provide more precise effect latencies in the next part.

### Prestimulus alpha power may spread spatiotemporally, and only occipital alpha power predicted perceptual behavior shortly before the stimulus onset

GLMM was used in various time windows across specified brain areas to establish effect latencies more precisely. In the time window of  $[-550, -450]$  ms, alpha power in the left parietal area showed a significant difference between hit and miss trials ( $p = 0.028$ , Fig. 4a left panel). The regression line indicates that corresponding prestimulus alpha power could predict subsequent hit rate in this brain area and time window (Fig. 4a right panel). Similarly, alpha power in the frontal, parietal, and occipital areas all could predict subsequent hit rates in different time windows (see corresponding  $p$ -values in supplemental Tables 1&2, Fig. 4b&c&d). It should be noted that we presented the statistical results in Fig. 4 without multiple comparisons correction. We also conducted multiple comparison corrections using the false discovery rate (FDR), and the adjusted  $p$ -values were demonstrated in supplemental Table 2. Most adjusted  $p$ -values in spatiotemporal windows ranged from 0.04 to 0.075, which showed marginal significance (See supplemental Table 2). Therefore, prestimulus alpha power may spread spatiotemporally from  $-450$  ms to  $-150$  ms, reflecting the dynamic engagement of different brain areas in the prestimulus stage. Furthermore, the frontal areas engaged relatively long before the stimulus onset  $[-450, -350]$  ms (Fig. 4b left), whereas only the occipital areas engaged shortly before the stimulus onset (Without FDR:  $p = 0.013$ & $0.008$ ,  $[-250, -150]$  ms, Fig. 4d left panel; FDR corrected:  $p = 0.04$ ,  $[-350, -250]$

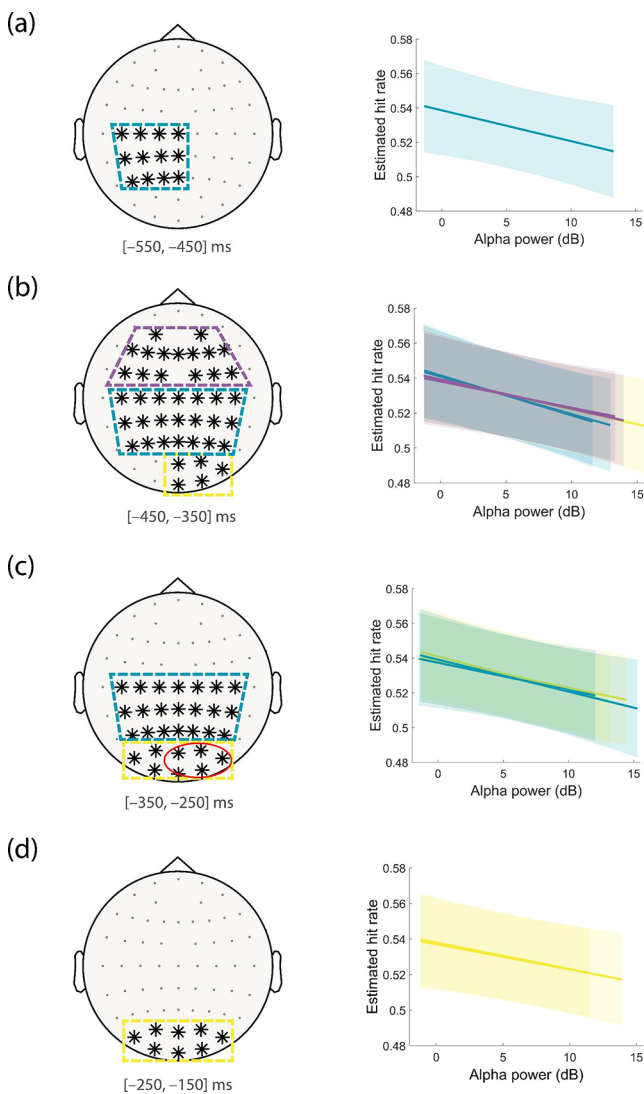
ms, see supplemental Tables 2 and red circle in Fig. 4c left panel). Thus, occipital alpha power directly affects perceptual performance, whereas alpha power over other areas may influence occipital alpha power.

### ERPs over the occipital area differentiated perceptual performance

GLMM results showed that prestimulus alpha power over the occipital area directly affected perceptual performance. To provide convergent evidence, we compared ERPs between hit and miss trials over the POz electrode. The brain reacted to the near-threshold stimulus in hit trials before the question mark, namely before 400 ms (Fig. 5a). The brain responded to the near-threshold stimulus in miss trials after the question mark, namely after 400 ms (Fig. 5b). Before the question mark's presentation, the brain showed a response difference between hit and miss trials (Fig. 5c). ERPs difference between hit and miss trials across the brain in  $[0, 400]$  ms showed that the brain reacted to the near-threshold stimulus differently before the question mark, mainly over the occipital area (Fig. 5d). All ERP results demonstrated that the brain started differentiating the near-threshold stimulus (namely hit/miss) from the occipital area.

### Frontal-occipital phase synchronization predicted the occipital alpha power in the prestimulus stage

After determining that occipital activity directly affects visual perception, we investigated whether frontal-occipital (Fz-POz) phase synchronization could predict the occipital alpha power in the prestimulus stage by using both univariate and multivariate methods. Fig. 2a shows the analysis pipeline. For the calculations of phase synchronization and phase difference, the time interval to analyze, i.e.  $[-450,$



**Fig. 4** Estimation of effect latencies in spatiotemporal windows through GLMM. **a** The left panel shows that alpha power in the left parietal area in the time window of  $[-550, -450]$  ms could predict subsequent hit rate. The right panel depicts the corresponding regression line. The shaded areas represent 95% confidence intervals. **b** The left panel shows that alpha power in the frontal, parietal, and left occipital areas in the time window of  $[-450, -350]$  ms could predict subsequent hit rate. The right panel depicts the corresponding regression lines. **c** The left panel shows that alpha power in the parietal areas in the time window of  $[-350, -250]$  ms could predict subsequent hit rate. The right panel depicts the corresponding regression lines. The red circle means that the right occipital alpha power corresponded to a significant adjusted  $p$ -value using FDR. **d** The left panel shows that alpha power in the occipital areas in the time window of  $[-250, -150]$  ms could predict subsequent hit rate. The right panel depicts the corresponding regression lines

$-150]$  ms, were chosen as the interval in which the whole brain areas started to be included in GLMM analysis. For the calculations of predicted alpha power, the interval was chosen to be  $[-250, -150]$  ms, which exclusively showed the difference shortly before the stimulus onset in GLMM

analysis. For the univariate analysis shown in Fig. 2b, the theta band presented a significantly smaller phase synchronization in the low-power ( $0.56 \pm 0.03$ ) than high-power conditions ( $0.58 \pm 0.02$ ), with  $t(29) = -2.63$ ,  $p = 0.01$ , corresponding to a Cohen's  $d$  effect size of  $d = -0.48$  (a small to medium effect). The alpha band also presented a significantly smaller phase synchronization in low-power ( $0.46 \pm 0.04$ ) than high-power conditions ( $0.50 \pm 0.06$ ), with  $t(29) = -5.35$ ,  $p < 0.001$ , corresponding to a Cohen's  $d$  effect size of  $d = -0.98$  (a large effect). The beta band did not show the difference with  $t(29) = -0.97$ ,  $p = 0.34$ . For the multivariate method, all the three frequency bands yielded the classification performance above the chance level (Fig. 2c), implying that frontal-occipital phase synchronization in  $[4, 30]$  Hz could predict the occipital alpha power.

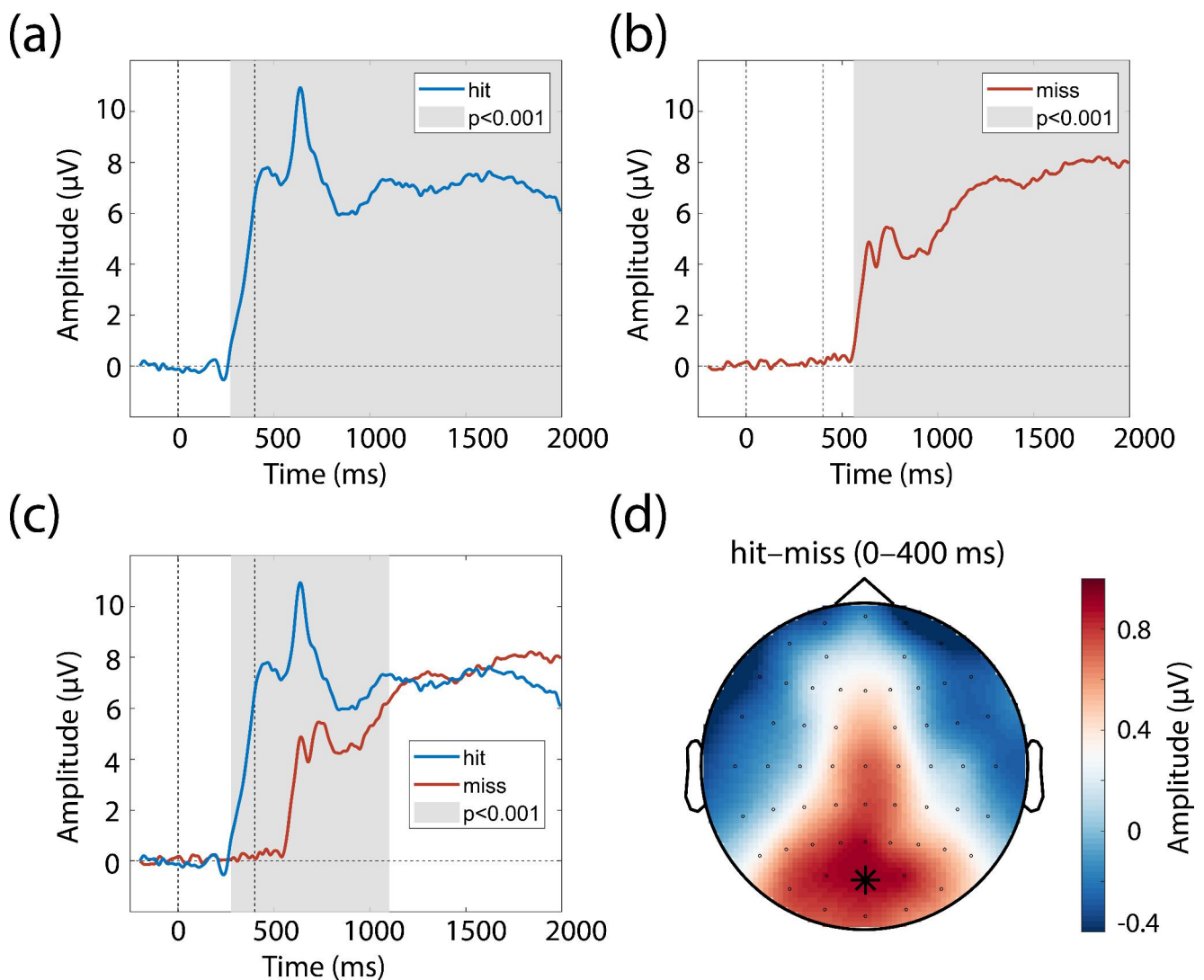
## Discussion

To investigate whether frontal-occipital phase synchronization could impact the occipital alpha power that affects the perceptual performance in a visual detection task, we used the general linear mixed model (GLMM) to yield a likely spatiotemporal distribution of global alpha power. Only the occipital power predicted the perceptual performance shortly before the stimulus onset, and ERPs further indicated that it was the occipital area where the perceptual performance difference occurred. After determining that the occipital alpha power directly impacts perceptual behavior, we supposed that the frontal activity would influence the occipital alpha power via phase synchronization. We used univariate and multivariate methods to analyze the frontal-occipital phase synchronization in three frequency bands. Both analyses showed that frontal-occipital phase synchronization predicted the occipital power in the prestimulus stage.

### It was the prestimulus alpha power over the occipital area, rather than other brain areas, to directly affect the perceptual performance

Regarding the prestimulus alpha power, we first replicated two analysis strategies that previous studies often used (Iemi et al. 2017; Limbach and Corballis 2016). The distribution difference of prestimulus alpha power probably came from two aspects. First, although the two strategies both analyzed prestimulus activity, they handled different types of values: one was alpha power, and the other was hit rate. Second, we conducted cluster-based permutation tests, which have limited capacity to establish precise effect latencies (Sassenhagen and Draschkow 2019). GLMM results further presented a likely spatiotemporal trajectory across





**Fig. 5** ERPs specifically over the occipital area and across the whole brain in hit and miss conditions. **a** ERPs of the hit trials over the POz electrode showed that the brain reacted to the near-threshold stimulus before the question mark, namely before 400 ms. The left vertical dash line labels the onset time of stimulus (0 ms). The right vertical dash line labels the presentation time of the question mark (400 ms). The grey patch labels the time interval with a significant difference after the cluster-based permutation test ( $\alpha = 0.05$ ). **b** ERPs of the miss trials over the POz electrode showed that the brain reacted to the near-threshold stimulus after the question mark, namely after 400 ms. **c** The ERPs difference indicated that the brain responded to the near-threshold differently between hit and miss trials before the question mark. **d** The ERPs difference across the whole brain between hit and miss trials was averaged before the question mark, namely in [0, 400] ms. The black asterisk labels the location of the POz electrode

whole brain areas. This trajectory spread from the frontal to occipital areas in  $[-450, -150]$  ms, which may relate to neural mechanisms of top-down control embodied in expectation, sustained attention, etc. When subjects expected that the stimulus would come very soon, the frontal area exerted cognitive control on the sensory areas with sustained attention, including motor and occipital areas (Fig. 4b) (Diepen et al. 2017; Sadaghiani and Kleinschmidt 2016). As a result, shortly before the stimulus onset, the occipital area became most active to prepare for the coming stimulus (Fig. 4c and d). Also, this frontal-occipital spread tendency may connect to other findings, such as the traveling waves (Halgren et

al. 2019) or feedback processing in the visual cortex (Van Kerkoerle et al. 2014). The electrocorticographic (ECoG) study showed that alpha waves might propagate from anterosuperior to posteroinferior cortex, implying a functional progression from higher-order to lower-order visual areas (Halgren et al. 2019). Although the traveling waves were beyond the scope of the current study, the likely frontal-occipital spread tendency may suggest a long-range traveling wave involving different stages of visual processing (Halgren et al. 2019; Van Kerkoerle et al. 2014). It should be noted that the above discussions were based on statistically unreliable results since  $p$  values in most spatiotemporal

windows could not survive multiple comparisons correction using FDR. However, as shown in supplemental Tables 2, most FDR-adjusted  $p$  values had marginal significance. Therefore, we argued that there was a high likelihood that the spatiotemporal pattern occurred.

GLMM results also suggested that the prestimulus alpha power over the occipital area directly predicted perceptual performance shortly before the stimulus onset. Being convergent, ERPs results showed that the occipital area reacted to the near-threshold stimulus differently between hit and miss trials before the question mark, which asked participants to respond by pressing buttons (Fig. 5). Hit trials responded to the near-threshold stimulus earlier than miss trials, demonstrating that the brain started to distinguish “see” or “not see” over the occipital area. Consequently, the (un)awareness of visual stimulus could be attributed to the occipital area covering the visual cortex, namely, the (un)awareness occurred in the very first step of the bottom-up process, the finding which complied with previous studies (Busch and VanRullen 2010; Koivisto and Revonsuo 2003; Koivisto et al. 2008; Förster et al. 2020). To conclude, ERP results convinced that the occipital activity took direct responsibility for stimulus (un)awareness. In contrast, other brain areas, including frontal and parietal, may impact occipital activity with top-down control (Sadaghiani and Kleinschmidt 2016; Clayton et al. 2015, 2018).

### Low-frequency frontal-occipital phase synchronization predicted the occipital power in the prestimulus stage

To investigate how the occipital alpha power was affected by top-down control from frontal activity, we used univariate and multivariate methods to analyze the effect of frontal-occipital phase synchronization. These two methods complement each other in this study. The univariate method informed the difference of averaged phase synchronization between two states (low/high power), while the multivariate method distinguished the patterns of phase difference (Haxby et al. 2014; Ritchie et al. 2019). The univariate analysis found that theta- and alpha-band phase synchronizations were lower in low-alpha-power compared with high-alpha-power conditions. This frequency band, [4, 13] Hz, could also be found in related studies, which mainly used univariate methods as well (Busch et al. 2009; VanRullen et al. 2011; Tomassini et al. 2017). The multivariate analysis employed a slightly different strategy of MVPA, which used to adopt either a hierarchical or an inter-subject way in the group-level analysis (Wang et al. 2020). Our study required decoding the time series of phase differences. Trials from all subjects were put together and classified into low- and high-power conditions. In this process, the information of

subjects was dropped out, and the general pattern of phase difference was identified. However, MVPA could distinguish information patterns that the brain may not use (Ritchie et al. 2019). To overcome this limitation, the univariate and multivariate methods handled information on phase difference differently: the univariate calculated phase lag index based on phase difference; the multivariate used phase difference directly. Considering the considerable overlap between results yielded by the two analysis methods, we argued that the multivariate method might succeed in identifying the latent neural patterns. Therefore, low-frequency (4–30 Hz) frontal-occipital phase synchronization predicted the occipital power in the prestimulus stage.

The phase synchronization results increased the complexity of related studies. We found that the higher theta- and alpha-band phase synchronization, the higher prestimulus alpha power in univariate analysis. However, it was tricky to recognize the extent of the underlying top-down control when connecting to some related psychological constructs. One of the top-down factors—sustained attention—could increase while the frontal-occipital alpha phase synchronization increased (Kong et al. 2017) or decreased (Liu et al. 2010). As a result, the extent of sustained attention was uncertain in our study’s different states of occipital excitability (high/low power). Future work could figure out top-down factors (e.g., sustained attention) by manipulating related behavioral variables.

Furthermore, our study suggested that the alpha band had the highest prediction effects in univariate and multivariate methods. However, a previous study showed that a phase reset between frontal eye fields (FEFs) and the occipital area at the beta frequency band rather than the alpha band might shape visual cortex activity and perception (Veniero et al. 2021). This inconsistency was probably derived from the different experiment designs, analysis methods, etc. This fact also reminded us that it should be careful to generalize findings across closely related experiments, such as visual detection and discrimination tasks. Even though they were often used to study the same cognition, some of their neural processes would differ.

### Limitations

There are some limitations to our study. First, we could not conclude that there was a spatiotemporal trajectory of prestimulus alpha power in Fig. 4. Although there was a high likelihood of the trajectory’s existence considering the marginal significance, the statistical validity can be achieved by increasing sample size, etc. Second, we did not use any computational models in this study. Though unnecessary due to our research questions, models such as Signal Detection Theory (Iemi et al. 2017) or Drift Diffusion Modelling

(Kloosterman et al. 2019) may help to uncover further the mechanisms of perceptual decision-making (Chen and Yang 2021; Vargas and Lauwereyns 2021). Third, when we said one specific EEG signature affected another signature or behavior, we could only infer that there was a correlation between them. To yield more firm conclusions, neuro-modulation methods have to be utilized to provide causal evidence. For example, transcranial alternating current stimulation (tACS) was proved to manipulate oscillations and thus be able to provide causal evidence between oscillations and cognition (Polanía et al. 2018). Future work could use tACS to modulate occipital alpha power and observe the perceptual behavior, or modulate the frontal-occipital phase synchronization and then see whether occipital alpha power is regulated.

## Conclusions

This study used the general linear mixed model (GLMM) and event-related potentials (ERPs) to pinpoint that the occipital activity directly affected visual perception. Using both the univariate and multivariate methods, we found that frontal-occipital phase synchronization in [4, 30] Hz could predict the prestimulus alpha power. Overall, our results showed that frontal-occipital phase synchronization underlying the top-down control could predict the occipital alpha power that directly affects perceptual decision-making.

**Supplementary Information** The online version contains supplementary material available at <https://doi.org/10.1007/s11571-022-09862-7>.

**Author contributions** Conceptualization: Jinwen Wei, Ziqing Yao, Gan Huang, Zhiguo Zhang; Methodology: Jinwen Wei, Ziqing Yao, Gan Huang, Zhiguo Zhang; Formal analysis and investigation: Jinwen Wei, Gan Huang, Linling Li, Zhen Liang, Li Zhang; Writing – original draft preparation: Jinwen Wei, Ziqing Yao; Writing – review and editing: Jinwen Wei, Ziqing Yao, Gan Huang, Linling Li, Zhen Liang, Li Zhang, Zhiguo Zhang; Funding acquisition: Zhiguo Zhang; Supervision: Zhiguo Zhang.

**Funding** This work was supported in part by [the Natural Science Foundation of Guangdong Province, China] (Grant number 2021A1515011152); in part by [the National Natural Science Foundation of China] (Grant number 81871443); in part by [Shenzhen-Hong Kong Institute of Brain Science-Shenzhen Fundamental Research Institutions] (Grant number 2021SHIBS0003); in part by [Shenzhen's Sanming Project of Medicine] (Grant number SZSM202111009); and in part by [Shenzhen Science and Technology Research and Development Fund for Sustainable Development Project] (Grant numbers KCXFZ20201221173400001, KCXFZ20201221173613036).

**Data and code availability** The study in this article earned Open Data and Open Materials badges for transparent practices. Data and materials for this study can be found at <https://osf.io/pvga6/>.

## Declarations

**Conflict of interest** The authors have no relevant financial or non-financial interests to disclose.

**Ethics approval** This study was approved by the ethics review board of Shenzhen University and informed consent was obtained from all the participants.

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